DS5003 Data Engineering Lab

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Dataset

You can use the following movie dataset `https://grouplens.org/datasets/movielens/`.

Problem Statement

1. You have to process the given .csv files from Popular Movies Datasets and need to store data in a database. (You may need to reorganize the data given in six files into different tables with slightly varying schemas to answer the following question.)

2. Access the data from the database to analyze it and visualize the outputs.

3. You need to create a dashboard to answer the following questions:

   (a) For a particular movie, show the top-5 tags based on the relevance scores of the tags for that movie.

   (b) Show overall and year-wise top 10 movies based on the average rating.

   (c) What are the most popular tags given for each genre.

   (d) For a particular movie, show the distribution of users' ratings.

   (e) Show the distribution of total ratings for movies.

   (f) For a particular year, show which genre released the highest number of movies.

   (g) Show genre-wise coverage of movies.

4. The database can be accessed in two roles: administrators and users. Users are allowed to modify only the rating of movies (if possible)

Technical aspects that must be present

- User login for the website (with roles if possible)
- Cleaning and preprocessing of data (if required)
- Visualizing the output
- User interaction

External tools and libraries can be used:

- For visualization and user input
- For website (if done)

- For any other tools we need to discuss and check, as they may depend on the approach of the teams

Deliverables

1. **Demo:** Here, you have to create a presentation/slides, highlighting important parts of your project. It must address the questions mentioned in the problem statement using the demo system you will build.

2. If you are using another dataset, please consider to share the problem statement with us. It must include at least seven exploration scenarios similar to the problem statement mentioned above.